BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: DIK-SSC-IX-2021(A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 12 | **Paper Code:** DIK-SSC-IX-2021(A)

Q1. Choose the correct answer from four possible answers (A,B,C,D) and fill the correct answer on MCQ's sheet separately.

Q1. Force needed to produce an acceleration of $20ms^{-2}$ in a ball of mass 0.5 kg is ._____

(A) 10N
(B) 15N
(C) 25N
(D) 100N

Q2. Colck wise torque is taken as _____

(A) Zero
(B) 1
(C) Positive
(D) Negative

Q3. The rate of doing work is called _____

(A) Work
(B) Energy
(C) Horse power
(D) Power

Q4. The transfer of heat by convection occurs easily in _____

(A) Gases
(B) Liquids
(C) Solids
(D) None

Q5. One horse power is equal to _____

(A) 476W
(B) 674W
(C) 746W
(D) 552W

Q6. The energy possessed by a body due to its motion is called _____ energy.

(A) Solar
(B) Heat
(C) Potential
(D) Kinetic

Q7. The unit used for pressure in weather map is _____

(A) Pascal (B) Atm

(C) Bar (D) N/m^2

Q8. The boiling point of water on Kelvin Scale is _____

(A) 100K (B) 373K

(C) 430K (D) 273K

Q9. Straight distance between two points is called_____

(A) Distance (B) Displacement

(C) Length (D) Motion

Q10. The famous book "Kitab-ul-Manazir" was written by:

(A) Ibnal Haitham (B) Yaqub Kindi

(C) Al-Beruni (D) Abdul Qadeer Khan

Q11. The S.I unit of resistance is_____

(A) Newton (B) Farad

(C) Ohm (D) Joule

Q12. Which of the following is a vector quantity?

(A) Power (B) Weight

(C) Speed (D) Volume

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 21 | **Paper Code:** DIK-SSC-IX-2021(A)

Q2. Attempt any 8 questions of the following:

(i) Define S.I base units. Enlist the names and symbols of base units.

(ii) Why a balloon filled with air move forward when its air is released?

(iii) Which material is more elastic steel or rubber and why?

(iv) A ball is thrown upward with an initial speed of 5m/s. What will be its speed when it returns to starting point?

(v) Differentiate between real and apparent expansions of liquids.

(vi) What is Pascal's principle? Explain it with example.

(vii) Why is the surface of a conveyor belt made rough?

(viii) State and explain law of universal gravitation,

(ix) Moon is attracted by the earth why it does not fall on earth?

(x) Define and explain power.

(xi) The temperature of a normal human body is $37^{o}C$. Find this temperature on the Kelvin Scale.

SECTION – C

Maximum Marks: 24

> Note: Attempt any 3 questions: Each carry 8 marks.

Q3.

(a) What is force? What are its unit? Distinguish between contact and non-contact forces.

(b) Calculate the mass of a body when a force of 700N produce an acceleration of $12.5ms^{-2}$

Q4.

(a) Prove that $S = vit + \frac{1}{2}at^2$ using graphical method.

(b) A golf ball that is initially traveling at 25m/s hits a sand trape and slows down with an acceleration of $-20m/s^2$. Find its displacement after two seconds.

Q5.

(a) Determine the mass of earth by applying law of universal gravitation.

(b) The maximum height of a typical human can jump is about 60cm by how much does the gravitational potential energy increase for 72kg person?

Q6.

(a) Define kinetic energy. Derive the expression used for kinetic energy.

(b) Write note on Anomalous expansion of water.